

CAN CIRCUIT DESIGN REVIEW

Production Readiness — Defect Report

Project: CAN GSM Controller Module | IC: TJA1042T | MCU: AT32L021C8T7

Date: April 29, 2026 | Classification: Engineering Internal

Executive Summary

This report documents all identified defects, omissions, and risks in the CAN bus interface section of the CONTROLLER_GSM schematic (Sheet 1). The analysis is conducted against production automotive/industrial standards including ISO 11898-2 (CAN physical layer), IEC 61000-4 (EMC immunity), and ISO 7637 (automotive transient immunity). Seven critical or major issues were found. The design, as currently drawn, is **not suitable for production deployment** and requires the corrections described herein before PCB fabrication.

Defect ID	Description	Severity	Status
CAN-D01	No Common-Mode Choke on CANH/CANL	CRITICAL	Open
CAN-D02	Bus Termination Strategy Undefined	CRITICAL	Open
CAN-D03	TVS Diodes Undersized for Automotive	MAJOR	Open
CAN-D04	No Decoupling Capacitors on VCC (U2)	MAJOR	Open
CAN-D05	STB Pin Hardwired to GND — Developer Error (must connect to MCU GPIO)	MAJOR	Open
CAN-D06	No Series Resistors on TXD/RXD Lines	MINOR	Open
CAN-D07	No Reverse Polarity / Surge Protection on +12V	MAJOR	Open

CRITICAL	Will cause field failure or non-compliance. Must be fixed before any production build.	MAJOR	High risk of intermittent failure in field. Fix before mass production.	MINOR	Best practice deviation. Recommended fix.
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Detailed Defect Analysis

Affected Net	CANH (J1 pin 6 → U2 pin 7) & CANL (J1 pin 5 → U2 pin 6)
Affected Parts	U2 (TJA1042T), J1 (BD-II-16M OBD Connector)
Standard Ref.	ISO 11898-2 §6.5; CISPR 25 Class 5; IEC 61000-4-6
Root Cause	CANH and CANL route directly from the OBD connector to the TJA1042T transceiver with no filtering element between them. A vehicle CAN bus is a high-noise environment where ignition switching, alternator ripple, and motor loads inject common-mode noise onto both bus lines simultaneously. Without a Common-Mode Choke (CMC), this noise passes directly into the transceiver and onto the MCU.
Risk / Failure Mode	1. False CAN frames — noise spikes cross the differential threshold causing phantom bit errors and dropped frames. 2. Transceiver damage — high-frequency common-mode voltages may exceed U2 absolute maximum ratings. 3. Regulatory non-compliance — radiated and conducted emissions will fail CISPR 25 automotive EMC testing.
Required Fix	Insert an automotive-grade AEC-Q200 qualified Common-Mode Choke (CMC) between the OBD connector and the TVS diodes. All candidates must meet: AEC-Q200 qualification, operating temp -40 degC to +125 degC, impedance ≥ 90 Ohm at 1 MHz (common-mode), rated current ≥ 0.5 A. Recommended automotive-grade options (select any one): (1) TDK ACT45B-510-2P-TL003 — 51 uH, 1 A, AEC-Q200, -40/+125 degC, 0805 package; (2) Murata DLW21SN900SQ2L — 900 Ohm at 100 MHz, 0.3 A, AEC-Q200, 0805 package; (3) Bourns SRF2012-900Y — 90 Ohm at 1 MHz, 0.5 A, AEC-Q200, 0805 package; (4) Wurth Elektronik 744232090 — 90 Ohm, 0.6 A, AEC-Q200, -40/+125 degC; (5) Vishay IHCM0603ABER900J5A — AEC-Q200, -55/+125 degC, CAN-optimised impedance profile. Place whichever is selected as close as possible to J1. Trace routing: J1 CANH/CANL → CMC → TVS → TJA1042T.
Correct Signal Path	OBD Connector → Common-Mode Choke (AEC-Q200) → TVS Clamp (AEC-Q101) → TJA1042T (pins 6 & 7)

Affected Net	CANH, CANL — split termination network via R2
Affected Parts	U2 (TJA1042T) SPLIT pin 5, R2 (ERA2AEB121X = 120 Ω) — must be removed
Standard Ref.	ISO 11898-2 §7.1 — Bus termination; SAE J1962 (OBD-II connector spec); CAN in Automation (CiA) 601
Confirmed Node Position	This PCB connects to the vehicle CAN bus via the OBD-II diagnostic port (J1, BD-II-16M). The OBD-II port is a STUB / TAP point off the existing vehicle CAN backbone — it is definitively a MID-BUS node. The vehicle manufacturer has already placed two 120 Ω termination resistors at the physical ends of the backbone (typically inside the ECU and one other major node such as the ABS or BCM controller). This module is NOT and cannot be a bus-end node in this topology.
Bus Topology	[ECU]---120 Ohm---[CAN Backbone CANH/CANL]---120 Ohm---[ABS/BCM] with OBD-II port as a short stub branching off the backbone, and this PCB hanging off that stub.
Root Cause	R2 (120 Ω split termination via TJA1042T SPLIT pin) is present and always active. With this module plugged into the OBD-II port, the vehicle CAN bus now has THREE termination points instead of the required two. The total bus differential impedance drops from 60 Ω (two 120 Ω in parallel) to approximately 40 Ω (three 120 Ω in parallel). This 33% impedance reduction overloads the bus drivers of every ECU on the network.
Risk / Failure Mode	1. Waveform distortion — reduced bus impedance causes CANH/CANL differential voltage to fall below the ISO 11898-2 minimum of 1.5 V in dominant state, causing bit errors across ALL nodes on the vehicle bus — not just this module. 2. Vehicle-wide CAN failure — the car's own ECUs (engine management, ABS, airbag, etc.) may lose communication with each other, triggering DTC fault codes and potentially disabling safety-critical systems. 3. Increased reflections — the extra impedance discontinuity at the OBD stub adds signal reflections that degrade eye-diagram margin at high bit rates. 4. Regulatory/liability exposure — corrupting a vehicle's CAN network violates UNECE R155 cybersecurity and functional safety obligations.
Required Fix	Remove R2 entirely from the schematic and BOM. Disconnect the SPLIT pin (U2 pin 5) — leave it floating or tie to GND through a 100 nF capacitor as recommended by the TJA1042T datasheet for unused SPLIT. This PCB must present only the passive high-impedance load of the TJA1042T receiver input (typically >10 k Ω in recessive state) to the vehicle bus. No termination resistor of any value should be present on this board.
Post-Fix Bus Topology	[ECU]---120 Ohm---[CAN Backbone]---120 Ohm---[ABS/BCM] OBD-II stub → This PCB (high-Z receiver input only, no termination). Total termination: 60 Ω differential — correct per ISO 11898-2.

Affected Parts	D1, D2 — ESDSLC3V3LB-TP (ESD suppressor, clamping voltage ~5 V, I _{pp} ~1 A)
Standard Ref.	ISO 7637-2 (automotive transient pulses); ISO 16750-2 (load dump)
Root Cause	The ESDSLC3V3LB-TP is a consumer-grade ESD protection diode rated for electrostatic discharge in handheld electronics (IEC 61000-4-2). It is not rated for the transient environment found on a vehicle 12 V bus. Automotive ISO 7637-2 pulse definitions: Pulse 1 — negative spike to -100 V (load disconnect); Pulse 2a — positive spike to +112 V (inductive load switching); Load dump (Pulse 5) — up to +87 V for 400 ms. The ESDSLC3V3LB-TP will be destroyed by any of these pulses.
Risk / Failure Mode	1. Diode destruction on first ignition cycle — field return on Day 1. 2. Once D1/D2 are damaged, CANH/CANL are unprotected — next transient kills U2. 3. A failed-short TVS will short CANH or CANL to GND, taking down the entire CAN bus.
Required Fix	Replace D1 and D2 with automotive-qualified CAN-specific TVS devices. All candidates must meet: AEC-Q101 qualification, operating temp -40 degC to +150 degC, standoff voltage >= 24 V, peak pulse power >= 200 W (ISO 7637-2 compliant). Recommended automotive-grade options (select any one for both D1 and D2): (1) PESD1CAN (Nexperia) — bidirectional CAN TVS, 58 V clamping, AEC-Q101, SOD-323 package; (2) SM24CAN (STMicroelectronics) — 24 V standoff, 200 W peak, AEC-Q101, DO-219AB; (3) NUP2202W1T2G (onsemi) — dual-line, 400 W peak, AEC-Q101, SC-88 package; (4) CDSOT23-CAN (Bourns) — CAN-specific, 400 W peak, AEC-Q101, SOT-23 package; (5) PRTR5V0U2X (Nexperia) — dual TVS, AEC-Q101, -40/+125 degC, SOT-363; (6) BZT52C24S-HE3 (Vishay) — 24 V zener TVS, AEC-Q101, SOD-323 package. Preferred choice for CAN: PESD1CAN or CDSOT23-CAN as they are purpose-designed for CAN bus protection topology (CANH to GND and CANL to GND simultaneously).
Note on Layout	TVS diodes must be placed as close as possible to J1, before the CMC in the signal path. Long traces between the connector and the TVS defeat the protection. Connect TVS GND directly to chassis/board GND with a short, wide trace or via.

Affected Parts	U2 (TJA1042T pin 3 VCC)
Standard Ref.	TJA1042T Datasheet Rev 4.0 §8 — Application Circuit; IPC-2221 §6.3 — Decoupling
Root Cause	The TJA1042T VCC pin (pin 3) is connected to +3.3 V with no local decoupling capacitors visible on the schematic. The TJA1042T datasheet explicitly requires a 100 nF ceramic decoupling capacitor placed within 1 mm of the VCC pin, with an optional 10 uF bulk capacitor. CAN transceivers switch their output drivers at high speed (up to 1 Mbit/s), causing large instantaneous current demands that cannot be met by remote bulk capacitance on the power rail.
Risk / Failure Mode	1. Supply voltage dip on each CAN bit transition — causes glitches on the transmitted waveform, increasing bit error rate. 2. Noise injection back onto the +3.3 V rail affecting the MCU (U1) and other 3.3 V logic. 3. Potential violation of TJA1042T VCC ripple specification causing erratic bus state transitions.
Required Fix	Add automotive-grade AEC-Q200 qualified decoupling capacitors physically close to U2 pin 3. All candidates must meet: AEC-Q200 qualification, X7R or C0G dielectric (do NOT use Y5V/Z5U — capacitance drops severely with temperature in automotive range), operating temp -40 degC to +125 degC minimum. For C_local1 (100 nF bypass, place within 1 mm of pin 3): (1) Murata GCM188R71E104KA57D — 100 nF, 25 V, X7R, AEC-Q200, 0402; (2) TDK C1005X7R1E104K050BB — 100 nF, 25 V, X7R, AEC-Q200, 0402; (3) Yageo CC0402KRX7R9BB104 — 100 nF, 50 V, X7R, AEC-Q200, 0402; (4) Vishay VJ0402X104KXACW1BC — 100 nF, 50 V, X7R, AEC-Q200, 0402. For C_local2 (10 uF bulk, place within 5 mm of pin 3): (1) Murata GRM21BR61A106KE19L — 10 uF, 10 V, X5R, AEC-Q200, 0805; (2) TDK C2012X5R1A106M125AC — 10 uF, 10 V, X5R, AEC-Q200, 0805; (3) Yageo CC0805KKX5R8BB106 — 10 uF, 25 V, X5R, AEC-Q200, 0805; (4) Samsung CL21A106KOQNNNE — 10 uF, 10 V, X5R, AEC-Q200, 0805. Route both capacitors with shortest possible path to the local GND plane. Do not share the GND via with other noisy circuits.

Affected Parts	U2 (TJA1042T pin 8 STB) — currently tied to GND
Standard Ref.	TJA1042T Datasheet Rev 4.0 §7.1 — Mode Control; ISO 11898-6 — CAN partial networking
■ Specification Error	This defect originates from an incorrect hardware specification provided by the software engineering team. The software engineer responsible for defining the transceiver interface requirements incorrectly specified that STB should be connected to GND. The hardware company implemented the design exactly as instructed and is not at fault. The correct requirement — which should have been specified from the outset — is that STB must be connected to a GPIO of the MCU to allow firmware-level control of transceiver mode and fault recovery. This is an internal specification correction and the hardware company will need to be issued a revised requirement document before the next build.
Root Cause	STB (standby) pin is permanently connected to GND, forcing the TJA1042T into Normal mode at all times. While this ensures the bus is always active, it removes all MCU-level control over the transceiver. The STB pin serves two critical production functions: (1) power management — the TJA1042T can enter low-power standby ($I_{cc} < 10 \text{ uA}$ vs 5 mA in normal mode) saving significant quiescent current on battery-powered vehicle systems; (2) fault recovery — toggling STB is the recommended method to reset a transceiver stuck in a bus-off or error-passive state without a full system power cycle.
Risk / Failure Mode	1. No sleep/wake capability — the module will draw full current even when the vehicle is parked, accelerating battery drain in telematics applications. 2. If the transceiver enters bus-off state (due to excessive errors), the only recovery path is a complete power cycle — not acceptable in production. 3. Inability to meet any CAN partial-networking or ISO 11898-6 requirements if specified by the vehicle OEM.
Required Fix	Disconnect STB from GND. Connect STB directly to a dedicated MCU GPIO (e.g., an available PA port on U1 AT32L021C8T7) as originally specified. Add an automotive-grade AEC-Q200 10 k Ω pull-down resistor from STB to GND to ensure Normal mode (STB LOW = active) during MCU power-up as safe default. Update firmware to: assert STB HIGH during sleep events; toggle STB LOW-HIGH-LOW to perform transceiver reset on bus-off detection.
Pull-down Resistor — Automotive-Grade Options (AEC-Q200, -40/+125 degC)	(1) Yageo RC0402FR-0710KL — 10 k Ω , 1%, AEC-Q200, 0402, 63 mW; (2) Vishay CRCW040210K0FKED — 10 k Ω , 1%, AEC-Q200, 0402, -55/+155 degC; (3) Panasonic ERJ-2RKF1002X — 10 k Ω , 1%, AEC-Q200, 0402, -55/+155 degC; (4) Bourns CR0402-FX-1002ELF — 10 k Ω , 1%, AEC-Q200, 0402; (5) Murata ERA-2AEB103X — 10 k Ω , 0.1%, AEC-Q200, 0402 (premium option). Any of these is acceptable. Select based on availability and procurement preference.

Affected Parts	U2 TXD (pin 1) → U1 PA2; U2 RXD (pin 4) → U1 PA3
Standard Ref.	TJA1042T Application Note AN10211 §4.2
Root Cause	The TXD and RXD signal lines between the AT32L021 MCU and the TJA1042T transceiver have no series resistors. Although these are short board-level traces, the MCU GPIO output impedance is very low (typically 20-40 Ω), which, combined with PCB trace capacitance and the transceiver input capacitance, can cause ringing on fast edge transitions.
Risk / Failure Mode	1. Ringing on TXD can cause the transceiver to see multiple edges per bit, generating spurious CAN bits. 2. If trace lengths exceed ~30 mm without resistors, impedance discontinuities become significant at CAN edge frequencies. 3. Without a series resistor on RXD, any ESD event on CANL/CANH that passes through U2 can directly stress the U1 RXD input pin.
Required Fix	Add automotive-grade AEC-Q200 qualified 33 Ω series resistors on both TXD and RXD lines, placed at the MCU side (U1 PA2 and PA3 pins). All candidates must meet: AEC-Q200 qualification, -40 degC to +125 degC, 1% tolerance. Recommended automotive-grade options (select any one for both resistors): (1) Yageo RC0402FR-0733RL — 33 Ω , 1%, AEC-Q200, 0402, 63 mW; (2) Vishay CRCW040233R0FKED — 33 Ω , 1%, AEC-Q200, 0402, -55/+155 degC; (3) Panasonic ERJ-2RKF33R0X — 33 Ω , 1%, AEC-Q200, 0402, -55/+155 degC; (4) Bourns CR0402-FX-330ELF — 33 Ω , 1%, AEC-Q200, 0402; (5) Murata ERA-2AEB330X — 33 Ω , 0.1%, AEC-Q200, 0402 (premium). This damps ringing without affecting signal integrity at CAN bit rates up to 1 Mbit/s.

CAN-D07	No Reverse Polarity or Surge Protection on +12V Input	MAJOR
Affected Parts	J1 (BD-II-16M) +12V input feed to power supply stage; affects CAN transceiver VCC supply chain	
Standard Ref.	ISO 7637-2 Pulse 1/2a; ISO 16750-2 §4.6 — Reverse polarity; SAE J1113-11	
Root Cause	The +12V line from the OBD connector enters the board directly with only a polyfuse (F1, 0ZCG0050AF2C — 500 mA). There is no reverse polarity protection diode and no transient suppressor on the 12 V rail. In a vehicle environment: 1. Jump-starting can reverse-bias the battery for a short period. 2. Load dump when alternator disconnects from battery generates positive transients up to +87 V lasting 400 ms. 3. Technician wiring errors during installation frequently apply reverse polarity. The polyfuse provides overcurrent protection only — it offers zero protection against voltage transients or reverse polarity.	
Risk / Failure Mode	1. Reverse polarity — all ICs on the board powered from +12V (including the CAN power chain) will be destroyed immediately. No recovery. 2. Load dump transient — voltage exceeds maximum ratings of the step-down regulator (U5 AP63205WU7, max VIN typically 32 V) and all downstream ICs. 3. Field return rate will be high if vehicles are serviced by third parties.	
Required Fix	Add two protection stages on the +12V input rail, both with automotive-grade components. STAGE A — Load Dump TVS Clamp (place immediately after F1 polyfuse): All candidates must be AEC-Q101, rated for ISO 7637-2 Pulse 5 (87 V, 400 ms), standoff ≥ 33 V, peak power ≥ 600 W. (1) Littelfuse P6KE36CA — 36 V standoff, 600 W, AEC-Q101, DO-204AC; (2) Vishay SMAJ36CA-E3/61 — 36 V, 400 W, AEC-Q101, SMA package; (3) STMicro SM6T36AY — 36 V standoff, 600 W, AEC-Q101, DO-214AA; (4) Nexperia BZW06-36B — 36 V, 600 W, AEC-Q101, DO-15; (5) onsemi SMCJ36AHE3 — 36 V, 1500 W, AEC-Q101, SMC package (highest rated). STAGE B — Reverse Polarity Protection (P-channel MOSFET, place after TVS): All candidates must be AEC-Q101, VDS ≥ -40 V, ID ≥ 2 A, $-40/+125$ degC. (1) Nexperia PMBT3906 configured as P-ch switch with gate resistor — AEC-Q101; (2) onsemi FDN306P — P-ch, -30 V, 2.4 A, AEC-Q101, SOT-23; (3) Vishay SI2307A — P-ch, -30 V, 2.7 A, AEC-Q101, SOT-23; (4) Diodes Inc. DMP3010LK3 — P-ch, -30 V, 3.8 A, AEC-Q101, SOT-23; (5) Infineon BSS84AKS — P-ch, -50 V, 0.13 A, AEC-Q101 (for lower current loads). Sequence: J1 → Polyfuse F1 → TVS Clamp (Stage A) → P-ch FET (Stage B) → Board +12V rail.	

Required CAN Interface Signal Path (Production-Correct)

The following table defines the mandatory signal path for the CAN bus interface in a production design. Every stage listed is required.

Stage	Component	Specification	Purpose
1	OBD Connector J1	BD-II-16M (existing)	Vehicle interface
2	Reverse Polarity + Surge (NEW)	P-ch FET + 36 V TVS on +12V	Protects entire board from ISO 7637 pulses

Stage	Component	Specification	Purpose
3	Automotive TVS — CANH/CANL (REPLACE D1, D2)	PESD1CAN or SM24CAN	CAN-specific transient clamp, AEC-Q101
4	Common-Mode Choke (NEW)	51 uH CMC, e.g. ACT45B-510	Suppresses common-mode EMI; mandatory for CISPR 25
5	TJA1042T Transceiver U2 (existing)	VCC decoupled (100 nF + 10 uF); STB → MCU GPIO; R2 REMOVED; SPLIT pin → 100 nF → GND	CAN physical layer; high-Z mid-bus node; no termination
6	Series Resistors TXD/RXD (NEW)	33 Ω, 0402, placed at MCU pins	Damps ringing; secondary ESD buffer
7	AT32L021C8T7 MCU U1 (existing)	STB GPIO configured in firmware	CAN controller; bus-off recovery via STB

Production Readiness Verdict

NOT APPROVED FOR PRODUCTION 2 Critical defects and 4 Major defects must be resolved before any production build. Design requires a full schematic revision and re-review by the hardware team. A prototype build for bench validation is acceptable only after CAN-D01 through CAN-D05 are corrected.

Upon completion of all schematic corrections, a second review must be conducted covering: updated BOM with AEC-Q101 qualified components, PCB layout review (trace impedance, component placement proximity), and EMC pre-compliance testing on the first article prototype.